

Manuscripts considered — hgsoc-tuboovarian-stage3c-r0-hrd-pen ding-idyq

Master inventory: every paper reviewed in this case

Generated 2026-06-24

Manuscripts considered — hgsoc-tuboovarian-stage3c-r0-hrd-per

Master inventory: 9 clinical (9 included, 0 considered & excluded) + 9 pre-clinical (6 included, 3 considered & excluded) + 4 additional trial publications/registrations. One entry per manuscript, sorted by year (newest first). Excluded entries are papers a Libby agent reviewed and chose NOT to feed to the board, with the exclusion reason captured.

Imunon/Imunon (2025)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** GEN-1 (IMNN-001) + carboplatin/paclitaxel +/- olaparib or niraparib maintenance
- **Indication / model:** newly-diagnosed advanced epithelial ovarian cancer (1L); —
- **Design:** randomized phase 3 IMNN-001 + standard NACT/adjuvant chemo +/- PARP maintenance (OVATION 3)
- **Effect size:** — — progression-free survival / overall survival
- **Case fit:** strong
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Thaker/Anwer (2024)

Reference: — · **Type:** clinical · **Inclusion:** included

- **Intervention:** GEN-1 (IMNN-001) + neoadjuvant carboplatin/paclitaxel
- **Indication / model:** newly-diagnosed FIGO stage III/IV epithelial ovarian cancer, neoadjuvant (neoadj); newly-diagnosed advanced epithelial ovarian cancer treated with neoadjuvant chemotherapy, the patient's on-study setting. Pre-specified benefit in the BRCAm/HRD and PARP-maintenance subgroups links the IL-12 axis to her HRD question.
- **Design:** randomized phase I/II (OVATION 2)
- **n:** 110
- **Effect size:** 0.74 HR — overall survival
- **Toxicities:** Intraperitoneal IL-12 immunogene therapy has been manageable in prior GEN-1 studies; characteristic effects are infusion-related and abdominal symptoms on the IP route. Topline phase 2; full per-term safety table not in the press/abstract material.
- **Case fit:** strong
- **Notes:** Randomized phase I/II underpinning her on-study GEN-1/IMNN-001 program in her exact neoadjuvant stage III/IV setting. Efficacy from SITC 2024 topline and the 2025 ASCO survival update (abstract 5516); no peer-reviewed primary paper or PMID yet, and no structured per-term AE table in the public topline material, so toxicities[] left empty pending full publication.

Imunon/Imunon (2024)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** GEN-1 (IMNN-001) + bevacizumab + neoadjuvant carboplatin/paclitaxel
- **Indication / model:** newly-diagnosed advanced ovarian / fallopian-tube / primary-peritoneal cancer (neoadj); —

- **Design:** phase I/II IMNN-001 + bevacizumab + neoadjuvant chemotherapy with second-look
- **Effect size:** — — safety and surgical/objective response (second-look)
- **Case fit:** strong
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Thaker PH/Anwer K (2024)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** GEN-1 (IMNN-001) + neoadjuvant carboplatin/paclitaxel
- **Indication / model:** newly-diagnosed FIGO stage III/IV ovarian cancer (neoadjuvant) (neoadj); —
- **Design:** randomized phase I/II IMNN-001 (GEN-1) + NACT vs NACT (OVATION 2)
- **n:** 110
- **Effect size:** — — progression-free survival / surgical response
- **Case fit:** strong
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Monk/Kristeleit (2022) — J Clin Oncol

Reference: [PMID 35658487](#) · **Type:** clinical · **Inclusion:** included

- **Intervention:** rucaparib
- **Indication / model:** newly-diagnosed advanced epithelial ovarian cancer (stage III-IV) in response after first-line platinum (maintenance); HRD-stratified; benefit in both the HRD-positive subgroup and the ITT. Germline-BRCA-wild-type status leaves her HRD placement dependent on the pending tumor assay.
- **Design:** double-blind randomized placebo-controlled phase 3 (ATHENA-MONO/GOG-3020/ENGOT-ov45)
- **n:** 538
- **Effect size:** 0.52 HR — progression-free survival (ITT)
- **Variance:** 95% CI 0.4–0.68; $p < 0.001$
- **Toxicities:** anemia $G \geq 3$ ($n=425$, 28.7%); neutropenia $G \geq 3$ ($n=425$, 14.6%); ALT/AST increased $G \geq 3$ ($n=425$, 10.6%)
- **Case fit:** partial
- **Notes:** Tests rucaparib in her exact stage III-IV first-line maintenance setting with HRD stratification. Approved off-label for 1L maintenance in some settings; her HRD-subgroup placement waits on the pending tumor assay.

Stegelmeier/Wootton (2022) — Biomedicines

Reference: [PMID 35203573](#) · **Type:** preclinical · **Inclusion:** included

- **Intervention:** bevacizumab (anti-VEGF)
- **Indication / model:** orthotopic syngeneic mouse model of end-stage epithelial ovarian carcinoma (ID8-derived)
- **Design:** Sustained VEGF neutralization restrains tumor angiogenesis and ascites-driving vascular permeability in disseminated ovarian carcinoma.
- **n:** multiple AAV arms vs control
- **Effect size:** moderate — Sustained bevacizumab expression extended survival in an orthotopic ovarian model and gave the largest single-agent survival gain among the antiangiogenic agents tested. The benefit held even

in the end-stage disease setting.

- **Variance:** vs AAV-control vector / untreated end-stage controls; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** Orthotopic syngeneic ovarian model is a good tissue match, but delivery is AAV-vectored continuous expression rather than the IV q3w antibody the patient receives, so dose kinetics differ. Supports the anti-angiogenic axis she is already on.

Liu JF/Bookman MA (2022) — J Clin Oncol

Reference:PMID 35290101 · **Type:** trial · **Inclusion:** included

- **Intervention:** olaparib + cediranib
- **Indication / model:** recurrent platinum-sensitive ovarian cancer (2L+); germline BRCA-stratified
- **Design:** open-label randomized 3-arm (NRG-GY004)
- **n:** 565
- **Effect size:** 0.86 — progression-free survival
- **Variance:** 95% CI 0.66–1.1; p=0.077
- **Case fit:** weak
- **Notes:** (toxicity table not extracted)

Banerjee/Friedlander (2021) — Gynecol Oncol

Reference:PMID 34412908 · **Type:** clinical · **Inclusion:** included

- **Intervention:** anastrozole
- **Indication / model:** ER and/or PR-positive recurrent/metastatic granulosa-cell / sex-cord stromal ovarian tumours (2L+); ER and/or PR-positive granulosa-cell / sex-cord stromal tumours, not high-grade serous. Included as aromatase-inhibitor mechanism precedent in ER-positive ovarian disease, not an enrolling match for her histology.
- **Design:** single-arm phase 2 (PARAGON/ANZGOG-0903 anastrozole cohort)
- **n:** 41
- **Effect size:** 78.9% CBR (2.6% PR, 76.3% SD) % — clinical benefit rate at 12 weeks
- **Toxicities:** arthralgia G $\geq$ 3 (2/54, 3.7%); hot flushes (n=54); fatigue G1-2 (n=54)
- **Case fit:** cross_tumor_only
- **Notes:** Mechanism precedent for aromatase inhibition in ER-positive ovarian disease, but in granulosa-cell/sex-cord stromal histology, not her high-grade serous tumour. High stable-disease rate supports endocrine therapy as a low-toxicity holding option, not a cytoreductive one. Safety denominator (54) is the full PARAGON anastrozole cohort.

González-Martín/Monk (2019) — N Engl J Med

Reference:PMID 31562799 · **Type:** clinical · **Inclusion:** included

- **Intervention:** niraparib
- **Indication / model:** newly-diagnosed advanced ovarian cancer at high risk of recurrence, in response after platinum-based chemotherapy (maintenance); all-comers regardless of BRCA or HRD status; HRD-positive subgroup pre-specified. Stage III with residual disease or any stage IV. Benefit larger in HRD-positive but

present across the ITT, so this option does not strictly require the pending tumor HRD result.

- **Design:** double-blind randomized placebo-controlled phase 3 (PRIMA/ENGOT-OV26/GOG-3012)
- **n:** 733
- **Effect size:** 0.62 HR — progression-free survival (ITT)
- **Variance:** 95% CI 0.5–0.76; $p < 0.001$
- **Toxicities:** anemia $G \geq 3$ ($n=484$, 31%); thrombocytopenia $G \geq 3$ ($n=484$, 28.7%); neutropenia $G \geq 3$ ($n=484$, 12.8%)
- **Case fit:** partial
- **Notes:** Only PARP option active across the ITT, so it does not gate on the pending HRD assay, though benefit is larger if HRD-positive. Cytopenia rates from the primary NEJM analysis; the 3.5-year update (Eur J Cancer 2023) reports higher cumulative grade 3+ thrombocytopenia (39.7%) but lower rates with the individualized starting dose.

Ray-Coquard/Harter (2019) — N Engl J Med

Reference: [PMID 31851799](#) · **Type:** clinical · **Inclusion:** included

- **Intervention:** olaparib + bevacizumab
- **Indication / model:** newly-diagnosed advanced high-grade serous/endometrioid ovarian cancer in response after first-line platinum plus bevacizumab (maintenance); all on bevacizumab; benefit concentrated in HRD-positive tumours (with or without BRCA mutation). She is already on bevacizumab, so this is the closest design match, but the olaparib add-on only pays off if the pending HRD score is positive.
- **Design:** double-blind randomized placebo-controlled phase 3, all patients on bevacizumab (PAOLA-1/ENGOT-ov25)
- **n:** 806
- **Effect size:** 0.59 HR — progression-free survival (ITT)
- **Variance:** 95% CI 0.49–0.72; $p < 0.001$
- **Toxicities:** hypertension $G \geq 3$ ($n=535$, 19%); anemia $G \geq 3$ ($n=535$, 17%); lymphopenia $G \geq 3$ ($n=535$, 7%); neutropenia $G \geq 3$ ($n=535$, 6%); fatigue/asthenia $G \geq 3$ ($n=535$, 5%)
- **Case fit:** partial
- **Notes:** Closest design match because every patient is on bevacizumab, which she already receives; the olaparib add-on benefit is HRD-gated. Final OS (Ray-Coquard, Ann Oncol 2023) shows a 5-year OS gain in HRD-positive tumours but no significant ITT OS benefit.

Moore/DiSilvestro (2018) — N Engl J Med

Reference: [PMID 30345884](#) · **Type:** clinical · **Inclusion:** included

- **Intervention:** olaparib
- **Indication / model:** newly-diagnosed advanced high-grade serous/endometrioid ovarian cancer with a BRCA mutation, in response after platinum-based chemotherapy (maintenance); germline or somatic BRCA1/2 mutation required; FIGO stage III-IV; complete or partial response to first-line platinum. Excludes BRCA-wild-type tumours, which is the gap for this germline-wild-type, somatic-BRCA-pending patient.
- **Design:** double-blind randomized placebo-controlled phase 3 (SOLO1/GOG-3004)
- **n:** 391
- **Effect size:** 0.3 HR — progression-free survival
- **Variance:** 95% CI 0.23–0.41; $p < 0.001$
- **Toxicities:** anemia $G \geq 3$ ($n=260$, 22%); neutropenia $G \geq 3$ ($n=260$, 8%); AML G5 ($n=260$, 1.2%); treatment

discontinuation for AE (n=260, 11.5%)

- **Case fit:** partial
- **Notes:** PFS HR from the 2018 primary analysis; median PFS, OS HR and toxicity drawn from the 5-year (Banerjee 2021) and 7-year (DiSilvestro JCO 2023) follow-ups via secondary sources. Eligibility hinges on the pending somatic-BRCA call, since germline BRCA is wild-type.

Heinzelmann-Schwarz/Vetter (2018) — Gynecol Oncol

Reference:PMID 29157627 · **Type:** clinical · **Inclusion:** included

- **Intervention:** letrozole
- **Indication / model:** estrogen-receptor-positive high-grade serous ovarian cancer, maintenance after first-line therapy (maintenance); ER-positive high-grade serous histology, the patient's exact biology. Low-burden oral therapy fitting her age band and stated preference; PR-negativity tempers expected magnitude.
- **Design:** prospective cohort with matched-control validation
- **n:** 80
- **Effect size:** 60% vs 38.5% recurrence-free at 24 mo % — recurrence-free survival at 24 months
- **Variance:** p=0.035
- **Toxicities:** Aromatase inhibitors are low-toxicity; class effects are hot flushes, arthralgia and fatigue. No high-grade hematologic or neuropathic toxicity, which fits her vetoes. Per-term AE breakdown not extractable from the cohort report.
- **Case fit:** partial
- **Notes:** Non-randomized cohort, so a weaker tier than the PARP and bevacizumab RCTs, but it targets her exact ER-positive high-grade serous biology with low-burden oral therapy. Source reports the recurrence-free signal without a structured per-term toxicity table; AE data not separable from this abstract/cohort report, so toxicities[] left empty per the class-effect summary.

Sun/Wang (2018) — Oncotarget

Reference:PMID 30647846 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** niraparib
- **Indication / model:** BRCA-mutant and BRCA-wild-type tumor xenografts (incl. intracranial TNBC); comparative PK across PARP inhibitors
- **Design:** Niraparib distributes into tumor tissue well above plasma, sustaining intratumoral PARP trapping and synthetic lethality in HR-deficient cells and producing growth inhibition even in BRCA-wild-type tumors.
- **n:** not reported per arm
- **Effect size:** moderate — Steady-state tumor exposure to niraparib was about 3.3x plasma, the inverse of olaparib's tumor-to-plasma ratio. Niraparib gave stronger tumor growth inhibition than olaparib at matched MTD, including in BRCA-wild-type models.
- **Variance:** vs vehicle; head-to-head vs olaparib at matched MTD; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** PK/efficacy comparison across tumor types rather than a dedicated ovarian study. The tumor-penetration advantage supports niraparib's clinical benefit in HRD-uncertain and BRCA-wild-type disease (PRIMA all-comers), which matters here because the patient's HRD score is pending.

Hirakawa/Mizunuma (2014) — J Ovarian Res

Reference:PMID 24410765 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** letrozole (aromatase inhibition)
- **Indication / model:** ovariectomized BALB/c nude mice, peritoneal carcinomatosis of ERalpha-positive OVCAR-3 vs ERalpha-low DISS
- **Design:** Aromatase inhibition lowers local estrogen, downregulating ERalpha and FOXP1 signaling and cutting tumor VEGF, so the antitumor effect tracks ERalpha expression and runs partly through reduced angiogenesis.
- **n:** n=10 mice/group
- **Effect size:** moderate — Letrozole prolonged survival in ERalpha-positive OVCAR-3 carcinomatosis but not in ERalpha-low DISS tumors. Tumor microvessel density fell to 5.8 vs 8.9 vessels/mm² and VEGF dropped, while caspase-3 apoptosis was unchanged.
- **Variance:** vs untreated control; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** ERalpha-dependence of the effect matches the patient's ER-positive biology, but OVCAR-3 is a high-ER line whereas her ER is weak-to-moderate intensity and PR-negative, tempering expected magnitude. Effect appears antiangiogenic/cytostatic rather than cytotoxic.

Arjaans/de Vries (2013) — Cancer Res

Reference:PMID 23580570 · **Type:** preclinical · **Inclusion:** excluded — Ovarian xenograft work, but the endpoint is bevacizumab-induced vessel normalization reducing antibody uptake (a drug-delivery caution), not antitumor efficacy against the patient's targetable VEGF axis. Mechanistically tangential to the maintenance decision.

- **Intervention:** bevacizumab (anti-VEGF)
- **Toxicities:** n/a (preclinical)
- **Notes:** Excluded: Ovarian xenograft work, but the endpoint is bevacizumab-induced vessel normalization reducing antibody uptake (a drug-delivery caution), not antitumor efficacy against the patient's targetable VEGF axis. Mechanistically tangential to the maintenance decision.

Burger/Liang (2011) — N Engl J Med

Reference:PMID 22204724 · **Type:** clinical · **Inclusion:** included

- **Intervention:** bevacizumab (concurrent + maintenance)
- **Indication / model:** newly-diagnosed advanced epithelial ovarian / fallopian-tube / primary-peritoneal cancer (stage III incompletely resected or stage IV) (1L); all-comers; the PFS benefit came only from the bevacizumab-throughout (concurrent + maintenance) arm, matching her current regimen. This is the direct evidence base for the anti-angiogenic axis she is already on.
- **Design:** double-blind randomized placebo-controlled 3-arm phase 3 (GOG-218)
- **n:** 1873
- **Effect size:** 0.717 HR — progression-free survival
- **Variance:** 95% CI 0.625–0.824; p=<0.001
- **Toxicities:** hypertension G≥2 (n=608, 22.9%); gastrointestinal perforation/fistula (n=608)
- **Case fit:** strong
- **Notes:** Establishes the bevacizumab maintenance regimen she is already receiving in her exact indication and

1L setting. PFS gain came only from the maintenance arm; final analysis (Tewari JCO 2019) showed no ITT OS benefit.

Perren/Oza (2011) — N Engl J Med

Reference:PMID 22204725 · **Type:** clinical · **Inclusion:** included

- **Intervention:** bevacizumab (concurrent + maintenance)
- **Indication / model:** newly-diagnosed ovarian cancer including high-risk stage IIIc/IV (1L); all-comers; largest benefit in the high-risk subgroup (stage IV or stage III with >1 cm residual disease), which overlaps her presentation before R0 resection.
- **Design:** open-label randomized phase 3 (ICON7)
- **n:** 1528
- **Effect size:** 0.81 HR — progression-free survival
- **Variance:** 95% CI 0.7–0.94; p=0.004
- **Toxicities:** hypertension G_≥2 (n=745, 18%)
- **Case fit:** strong
- **Notes:** Open-label, so RoB above SOLO1/PAOLA-1. Reinforces bevacizumab maintenance in her 1L setting with the largest gain in the high-risk subgroup she matched pre-resection. OS update (Oza Lancet Oncol 2015) confined the survival gain to that subgroup.

Kortmann/Gilks (2011) — Clin Cancer Res

Reference:PMID 21097693 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** olaparib
- **Indication / model:** patient-derived ovarian serous carcinoma tissue xenograft, BRCA2 germline-mutated (vs a BRCA-proficient serous xenograft)
- **Design:** Olaparib exploits the homologous-recombination defect of BRCA2-mutant tumor cells, and carboplatin adds interstrand crosslinks that the same defective repair machinery cannot resolve.
- **n:** engraftment-limited cohorts (single PDX per genotype)
- **Effect size:** strong — Olaparib inhibited growth of the BRCA2-mutant ovarian PDX, and the carboplatin combination either eradicated tumors or left only scar tissue on histology. A BRCA-proficient serous xenograft showed no olaparib effect, fixing the response to the BRCA2 defect.
- **Variance:** vs vehicle; plus BRCA-proficient serous xenograft as specificity control; translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Closest preclinical match to this tumor type: a BRCA2-mutant ovarian serous PDX with a proficient control. Single PDX per genotype limits generalization. Case match is partial because the patient's germline BRCA is wild-type and her tumor HRD/somatic-BRCA status is still pending.

Drew/Curtin (2011) — J Natl Cancer Inst

Reference:PMID 21183737 · **Type:** preclinical · **Inclusion:** excluded — Rucaparib (AG-014699/CO-338) preclinical activity in BRCA-mutant/HR-deficient xenografts duplicates the PARP synthetic-lethal mechanism already covered by the olaparib and niraparib rows. Kept as audit trail; not separately synthesized.

- **Intervention:** rucaparib

- **Toxicities:** n/a (preclinical)
- **Notes:** Excluded: Rucaparib (AG-014699/CO-338) preclinical activity in BRCA-mutant/HR-deficient xenografts duplicates the PARP synthetic-lethal mechanism already covered by the olaparib and niraparib rows. Kept as audit trail; not separately synthesized.

Bryant/Helleday (2005) — Nature

Reference:PMID 15829966 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** PARP inhibition (olaparib class; synthetic lethality)
- **Indication / model:** BRCA2-deficient and BRCA2-complemented isogenic cell lines (V-C8 hamster, human); clonogenic survival
- **Design:** PARP inhibition leaves unrepaired single-strand breaks that collapse replication forks into double-strand breaks; cells lacking BRCA2 cannot repair these by homologous recombination and die.
- **n:** n=3 biological replicates (representative)
- **Effect size:** strong — BRCA2-deficient cells were several orders of magnitude more sensitive to PARP inhibition than their BRCA2-proficient counterparts, with no DNA-damaging agent added. This was one of the two 2005 papers that defined the BRCA-PARP synthetic-lethal interaction.
- **Variance:** vs BRCA2-proficient isogenic counterpart; translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** In-vitro mechanism paper, not an ovarian model. It underpins the entire PARP-inhibitor maintenance rationale that gates this case, which is why it is kept despite no in-vivo ovarian arm. Relevance to this patient is conditional on the pending tumor HRD/somatic-BRCA result.

Farmer/Ashworth (2005) — Nature

Reference:PMID 15829967 · **Type:** preclinical · **Inclusion:** excluded — Companion 2005 mechanism paper to Bryant et al.; same BRCA-PARP synthetic-lethal finding in isogenic cell lines. Logged as the second co-discovery but not separately synthesized to avoid double-counting the same in-vitro concept.

- **Intervention:** PARP inhibition (olaparib class; synthetic lethality)
- **Toxicities:** n/a (preclinical)
- **Notes:** Excluded: Companion 2005 mechanism paper to Bryant et al.; same BRCA-PARP synthetic-lethal finding in isogenic cell lines. Logged as the second co-discovery but not separately synthesized to avoid double-counting the same in-vitro concept.

Yockman/Kim (2003) — J Control Release

Reference:PMID 12618034 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** GEN-1 / IMNN-001 (IL-12 plasmid lipopolymer)
- **Indication / model:** CT-26 subcutaneous syngeneic tumors, BALB/c mice (water-soluble lipopolymer / IL-12 plasmid)
- **Design:** A polymer-condensed IL-12 plasmid produces local, sustained IL-12 with limited systemic spillover, driving IFN-gamma and a cytotoxic-T-cell response against the tumor.
- **n:** tumor-bearing mice, multiple treatment arms
- **Effect size:** moderate — Repeated lipopolymer/IL-12 dosing at 4-day intervals gave maximal tumor

suppression and survival, beating less frequent dosing. The formulation kept IL-12 local and limited systemic distribution, the design rationale carried into GEN-1/IMNN-001.

- **Variance:** vs untreated / vehicle (lipopolymer without IL-12 plasmid); translatability: low
- **Toxicities:** n/a (preclinical)
- **Case fit:** cross_tumor_only
- **Notes:** Foundational lipopolymer/IL-12 proof-of-concept for the GEN-1 platform, but in a colon (CT-26) subcutaneous model, not ovarian. Later murine pmIL-12/PPC ID8 ovarian work (intraperitoneal, IL-12/IFN-gamma induction with reduced ascitic VEGF) extends this to the patient's tissue but is reported mainly in review/secondary form. Cross-tumor only.